

STAT 611 Class Project

Table of Contents:

Page 1: Introduction to the problem and the objective at hand

Page 1: Exploratory Data Analysis and Data Preprocessing

Page 2: Variable Selection and Model Development

Page 5: Final Model evaluation/interpretation and validity

Introduction

Before analyzing the data, it is crucial to establish whether the primary objective is inferential or predictive. The prompt indicates that "vehicles from the same production year were collected in order to understand the relationship between the price," which suggests that the analysis is inferential in nature. This implies the goal is not to forecast future outcomes, but rather to obtain accurate coefficient estimates that help us understand the relationships between variables—specifically, how they influence the dependent variable and whether they are statistically significant.

In addition to identifying statistically significant predictors, other inferential goals include estimating the true population values of the coefficients and evaluating the magnitude of their impact on the dependent variable. All modeling choices—formulas, assumptions, and methodological decisions—were guided by this inferential framework.

It is also important to clarify why the Dealer Price variable was excluded from the analysis. First, its inclusion was deemed unnecessary due to its high variability, which is likely driven by subjective factors such as dealer location, pricing strategy, and other unobserved variables that are not captured in the dataset. Second, the response variable MSRP was sufficient for the analysis; it presented no modeling issues and is theoretically more reliable. Manufacturers typically base MSRP on known production costs and market considerations prior to release, which enhances its consistency as a pricing reference.

EDA/ Data Preprocessing

Before fitting a model, it is essential to develop a strong understanding of the dataset we are working with. Upon reviewing the Google Sheet, it becomes clear that the dataset is heavily numerical, with only two categorical variables.

A closer inspection reveals that several observations (rows) contain missing values across certain columns (variables). Naturally, the first instinct is to address these missing values through some form of imputation. However, based on my current experience and skillset, I do not possess advanced tools for performing imputation in a way that I find methodologically satisfying. The only techniques that come to mind are simple imputation methods, such as mean or median substitution, applied based on the distribution of each numerical column.

My primary concern with this approach lies in its underlying mechanism: it replaces missing values with a constant, either the mean or the median. While this may seem like a practical solution, when the objective is to produce accurate parameter estimates, this method can be problematic. Filling multiple observations with the same constant can distort hypothesis testing and shift parameter estimates toward the central tendency. Moreover, it can artificially narrow confidence intervals and lead to misleading

p-values, potentially resulting in false positives regarding statistical significance. As a result, I ultimately chose to leave the missing values unaltered.

Moving on to descriptive statistics, the numerical variables appear to be fairly symmetrical. This is evident in the close proximity between their means and medians, suggesting minimal skewness. One exception, however, is MSRP, which is not symmetrical. The mean and median diverge noticeably, and it exhibits a relatively high variance. This is a notable characteristic that will become more relevant later in the analysis.

As for the categorical variables, they are highly imbalanced. For instance, the Vehicle Type variable is overwhelmingly dominated by Sedan, while Drive Wheels is somewhat more balanced but still skewed, with most vehicles being FWD.

One of the more nuanced challenges in this project was determining how to treat the Number of Cylinders variable—whether as categorical or numerical. Conceptually, I view it as categorical within the context of model interpretation. However, as will be seen later, it functions more effectively as a numerical variable in this particular analysis. For now, I treat it as categorical because it's important to note that the variable contains exactly eight unique values. Despite this range, only three of them—4, 6, and 8 cylinders—appear frequently. The remaining categories have very few observations, often fewer than seven each, making the distribution highly imbalanced.

Having now established the structure and characteristics of the dataset, we can proceed to the model-building phase of the report.

Variable Selection and Model Development

The first step that came to mind was to build a baseline model, which would serve as a reference point for further improvements. As mentioned earlier in the Exploratory Data Analysis (EDA) section, one challenge I encountered involved deciding whether to treat Number of Cylinders as a categorical or numerical variable.

The number of Cylinders was initially treated as a categorical variable with k levels, encoded using $k-1$ dummy variables following standard practice. This approach was implemented using PROC GLM and manually adjusted for use in PROC REG. This follows the typical constraint that a full set of k dummy variables is only used when the levels sum to 1, which was not the case here, especially considering the presence of missing values in some observations.

I refer to the baseline model using Number of Cylinders as a categorical variable as the three-indicator variable model. Fitting this model using PROC GLM posed no issues.

However, a limitation arose when using PROC REG, which does not natively support classification syntax. As a result, I manually encoded the categorical variables into dummy variables, again using k–1 encoding. VetyA–VetyE was to describe vehicle type and DrwhA/DrwhB was to describe drive wheels

When comparing this three-indicator model to a simpler two-indicator model—which includes only Vehicle Type and Drive Wheels as categorical variables—the former performed slightly better, with a marginally higher R^2 and adjusted R^2 (by approximately 0.10). While I initially preferred the concept of the three-indicator model, I ultimately chose to treat Number of Cylinders as a numerical continuous variable.

This decision was driven by the fact that only a few vehicles had uncommon cylinder counts such as 3, 10, or 12. Modeling it as a continuous variable allowed me to measure the incremental impact of a one-cylinder change on a vehicle's value. Although I lack deep domain expertise in automotive engineering, my research suggested that the number of cylinders significantly affects a car's design and performance. This supported the rationale to model it numerically, at least from a practical, appearance-based perspective.

Therefore, I adopted the two-indicator model structure, treating Number of Cylinders as a continuous predictor. Upon fitting this revised baseline model with all variables, the model produced an R^2 of 0.7631 and an adjusted R^2 of 0.7533, indicating a strong fit. However, residual analysis revealed a notable outlier that deviated from the general trend, which warrants further investigation.

At this stage of the analysis, the focus shifts to identifying influential and high-leverage points in the baseline model. The goal is to improve model reliability and ensure that no individual data point unduly influences the estimation of parameters.

To identify such points, I evaluated three key diagnostic measures: Cook's D, R-squared, and the Covariance Ratio (CovRatio). A data point was flagged as influential if it met the following criteria simultaneously: Cook's D was greater than 1, the CovRatio fell outside the acceptable range of approximately 0.84 to 1.16 (based on the formula $1 \pm 3(p/n)$), and the R-student was greater than or equal to 3.

Observation 288 met all of these thresholds and was identified as highly influential. It had a CovRatio of 0.0004, indicating a significant reduction in model precision. CovRatio outside $1 \pm 3(p/n)$ implies high sensitivity of the model's covariance matrix to the observation. The R-student value of 13.34, and a Cook's D value exceeding 1. These indicators clearly showed that this data point was distorting the model.

Further examination of Observation 288 revealed an MSRP of 142,465, which is more than three standard deviations above the mean MSRP of 32,774.85, with a standard

deviation of 19,431.72. Its horsepower value was approximately 477, which also deviated substantially from the mean of 215.89 and the standard deviation of 71.8. Since the observation deviated significantly in two independent variables—MSRP and horsepower—it was determined to be unrepresentative of the general dataset and a likely source of distortion in model estimation.

Given that the number of cylinders for this observation belonged to a common category, such as 4 or 6, removing it did not compromise the integrity of less frequent cylinder categories like 10 or 12. Therefore, I decided to remove it from the dataset.

A similar evaluation led to the removal of Observation 231. Although its Cook's D value was 0.039, which is below the commonly used threshold of 1, its CovRatio was 0.2339 and its R-student was 5.3864. Despite not meeting the Cook's D threshold, the combined values of the CovRatio and R-student were concerning enough to warrant a deeper review. Upon closer inspection, the observation was found to deviate enough from the rest of the data that I chose to remove it to protect the integrity of the model.

In an effort to improve the model as much as possible, I proceeded to the variable selection phase. However, I recognized a major concern with this process: evaluation metrics such as Mallows' Cp, the F-test, and R^2 can be affected by the presence of multicollinearity. Because of this, I first conducted a thorough multicollinearity diagnostic before proceeding with selection techniques.

Examining the Variance Inflation Factors (VIFs), I found that three variables had values exceeding 10, which typically indicates multicollinearity. These variables were VetyA, MPG Highway, and MPG City. I then examined the eigenvalues from the collinearity diagnostics, focusing on those closest to zero. Specifically, I investigated the lowest eigenvalue to assess variance proportions. The recommended threshold from the reference text suggests that multicollinearity should be investigated when the variance proportion associated with a small eigenvalue exceeds 30. The largest variance proportion observed was 14.86 percent, which does not surpass this threshold. Based solely on VIF, multicollinearity was confirmed. However, the eigenvalue-based method suggested a milder, perhaps moderate, level of multicollinearity.

Despite the relatively inconclusive eigenvalue analysis, I chose to address multicollinearity to prevent any potential distortion of coefficient estimates. My initial solution was to apply Ridge Regression, using a penalty term $k=0.04$, selected after examining the Ridge Trace Plot over the interval 0 to 0.5 in increments of 0.01. This approach successfully reduced multicollinearity, lowered the VIFs, and shrank the coefficients by incorporating a bias term into the model. However, due to my limited experience and despite reviewing the course text, I could not find an appropriate diagnostic metric, such as R^2 or MSE, that would allow for proper interpretation of the Ridge model's effectiveness.

I then attempted Principal Component Regression (PCR), but once again encountered the same issue: the lack of a clear and interpretable performance diagnostic. Due to limited diagnostic interpretability, these models were excluded from final consideration.

Instead, I turned to a more straightforward solution and dropped the variable with the highest VIF value—MPG Highway, which had a VIF of 18.18. This decision had a positive effect, stabilizing the model such that all remaining variables had VIFs below 10. With multicollinearity addressed, I resumed the variable selection process.

The next step involved deciding on the appropriate selection criteria. Since my objective is to obtain the most accurate and interpretable coefficient estimates, I opted to use the F-test. Specifically, I retained all variables with a p-value below 0.05 and discarded the rest, as this threshold identifies those predictors that significantly contribute to the model.

I also considered which selection strategy to apply—forward selection, backward elimination, or stepwise. After some reflection, I chose stepwise selection, as it strikes a balance between forward and backward methods. Stepwise regression allows the model to add variables that significantly improve the fit, while simultaneously evaluating whether any of the existing variables should be removed, given the presence of others. In my view, this makes it the most robust and balanced approach of the three. Before I fit the variable selection process, I wanted to incorporate the presence of an interaction term to see if the horsepower changes along with vehicle type and another of horsepower with Drive wheels.

The final model obtained through this process is shown in Figure A1. I then refit the model using the selected variables in PROC GLM. After fitting the model, I wanted to reassess it using the Partial F-test, as it evaluates whether a specific coefficient contributes significantly to the model while controlling for all other variables already included. If the p-value is less than 0.05, the variable is considered significant; otherwise, it does not contribute meaningfully to the model.

The results of the Partial F-tests are shown in Figure A2. The p-value for Wheelbase was 0.0884, and for the interaction term Horsepower*Drive_Wheels, the p-value was 0.0598. Since both values exceed the 0.05 significance threshold, I decided to drop these two terms from the model.

This led to the revised model shown in Figure A3, where all variables retained have Partial F-test p-values less than 0.05. I consider this to be my final model and will now move on to conduct an in-depth analysis of its results.

Final Model evaluation/interpretation and validity

Having obtained our variable of interest, addressed multicollinearity, and assessed influence and leverage points, I now move on to an in-depth analysis of the model. Personally, I always begin with model diagnostic charts, particularly residual analysis, to examine whether the model satisfies the assumptions of linear regression—namely, that errors have a mean centered around zero and exhibit constant variance (σ^2).

Starting with the residual percentile bar graph, we observe that the residuals are generally centered around zero and appear to be normally distributed. However, the shape of the residual spread resembled an unusually wide funnel, suggesting non-constant variance. After experimenting with the dependent variable (MSRP), I applied a log transformation. This adjustment improved the residual distribution, as shown in Figure B2, correcting the funnel-like pattern.

In addition to improving model assumptions, the log transformation also led to a noticeable increase in model performance. The R^2 and adjusted R^2 improved from approximately 0.80 (as shown in Figure B3) to around 0.84 (Figure B4), which marks a substantial improvement. The similarity between R^2 and adjusted R^2 suggests that all regressor variables contribute meaningfully to the model, with minimal impact from overfitting or irrelevant predictors.

Looking at the residual quantile plot, we see that the residuals are tightly clustered around the reference line, which reinforces that the residuals have a mean near zero, aligning with the assumption of normality. In the residual versus predicted values plot (Figure B2), the residuals display a random scatter with no clear pattern, indicating homoscedasticity—consistent variance across all levels of the predicted values. Together, these visual diagnostics confirm that the model meets the key assumptions required for multiple linear regression.

Moving to the ANOVA table, we observe a highly significant F-test with a p-value less than 0.05. This confirms that at least one of the regressors has a statistically significant impact on the dependent variable. Another key finding is the extremely low Root Mean Square Error (RMSE) of 0.189, which is roughly half the standard deviation of the log-transformed MSRP. This level of precision is impressive and supports the reliability of the model's predictions. These results are presented in Figure B5.

As stated in the previous section, I also reassessed the Partial F-tests to determine whether each individual variable contributes significantly when controlling for the others. The results reaffirm that all variables in the final model have p-values less than 0.05, indicating that each regressor is statistically significant in the presence of the others.

Looking at the ANCOVA table in Figure B7, we can examine the 95 percent confidence intervals for the coefficient estimates. These intervals help us understand not

only the direction of the effect but also the magnitude of change, giving us a better interpretation of how each variable characteristic influences car price.

One particularly interesting observation is that the p-value for the regression coefficient corresponding to vehicle type: Sedan is the only one among the vehicle types that contributes significantly to the model, as it has a p-value less than 0.05. The other vehicle types have p-values above 0.05, indicating that they do not significantly impact the dependent variable. Similarly, within the Drive Wheels variable, only FWD (Front-Wheel Drive) contributes significantly to the model, while AWD (All-Wheel Drive) does not, based on its p-value.

When looking at the interaction terms between Horsepower and Vehicle Type, it becomes clear that the interaction effects for Horsepower*Minivan and Horsepower*Pickup are not statistically significant. However, it's important to emphasize that interaction terms require the inclusion of all relevant indicator variables. Even if certain individual terms within the interaction are not significant, they must be retained if at least one of the related interaction terms contributes significantly to the model.

In conclusion, after developing our model with an inferential framework, we arrived at a final model that not only provides strong and interpretable coefficient estimates but also demonstrates solid predictive strength based on the available data. This balance between statistical interpretability and predictive performance reflects the overall robustness of the modeling process.

Figure A1:

Stepwise Selection Summary							
Step	Effect Entered	Effect Removed	Number Effects In	Number Parms In	Adjusted R-Square	F Value	Pr > F
0	Intercept		1	1	0.0000	0.00	1.0000
1	Horsepowe*Drive Whee		2	4	0.7190	343.84	<.0001
2	Horsepowe*Vehicle Ty		3	9	0.7472	9.91	<.0001
3	Weight		4	10	0.7641	29.23	<.0001
4	Width		5	11	0.7839	36.93	<.0001
5	MPG - City		6	12	0.7914	15.19	0.0001
6	Vehicle Type		7	17	0.7948	2.28	0.0466
7	Wheelbase		8	18	0.7967*	4.70	0.0308
* Optimal Value of Criterion							

Selection stopped because the candidate for entry has SLE > 0.05 and the candidate for removal has SLS < 0.05.

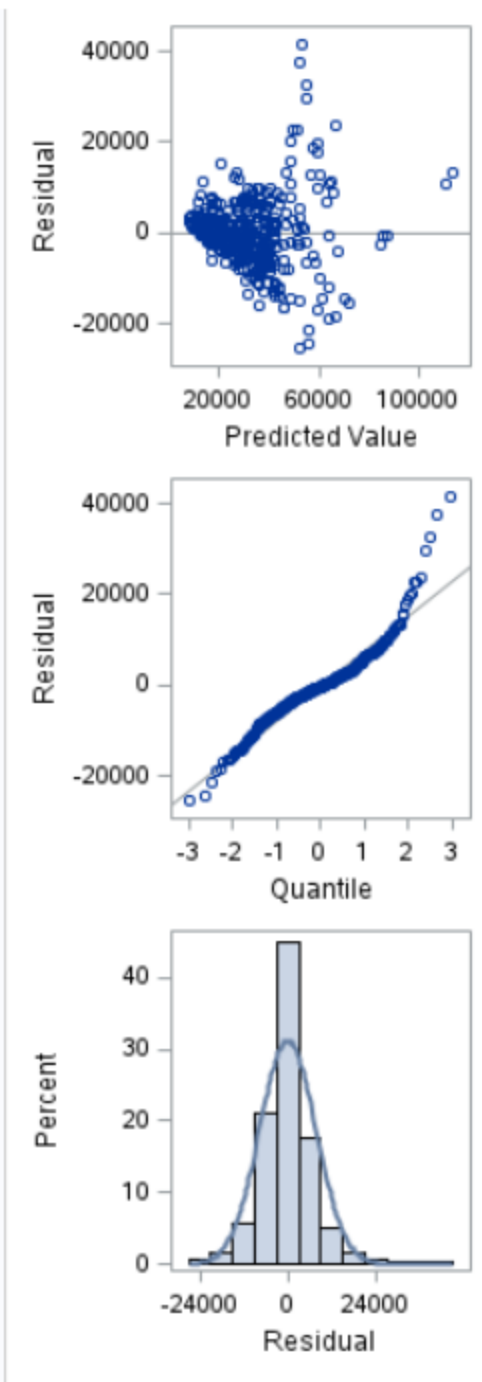
Figure A2:

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Vehicle Type	5	0.48433543	0.09686709	2.74	0.0190
Drive Wheels	2	0.45070904	0.22535452	6.37	0.0019
Horsepowe*Vehicle Ty	5	0.79960044	0.15992009	4.52	0.0005
Width	1	0.95760082	0.95760082	27.08	<.0001
Weight	1	4.29797612	4.29797612	121.56	<.0001
MPG - City	1	0.44300011	0.44300011	12.53	0.0004
Horsepower	1	1.41690897	1.41690897	40.07	<.0001
Horsepowe*Drive Whee	2	0.20069696	0.10034848	2.84	0.0598
Wheelbase	1	0.10317624	0.10317624	2.92	0.0884

FigureA3:

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Vehicle Type	5	0.69954919	0.13990984	3.90	0.0018
Drive Wheels	2	0.83218577	0.41609288	11.60	<.0001
Horsepower*Vehicle Ty	5	1.11548689	0.22309738	6.22	<.0001
Width	1	1.21156834	1.21156834	33.79	<.0001
Weight	1	4.40561187	4.40561187	122.86	<.0001
MPG - City	1	0.31025628	0.31025628	8.65	0.0035
Horsepower	1	1.31248942	1.31248942	36.60	<.0001

Figure B1:



Observations	403
Parameters	17
Error DF	386
MSE	6.12E7
R-Square	0.8005
Adj R-Square	0.7922

Figure B3:

Observations	403
Parameters	17
Error DF	386
MSE	0.0359
R-Square	0.8453
Adj R-Square	0.8389

Figure B4:

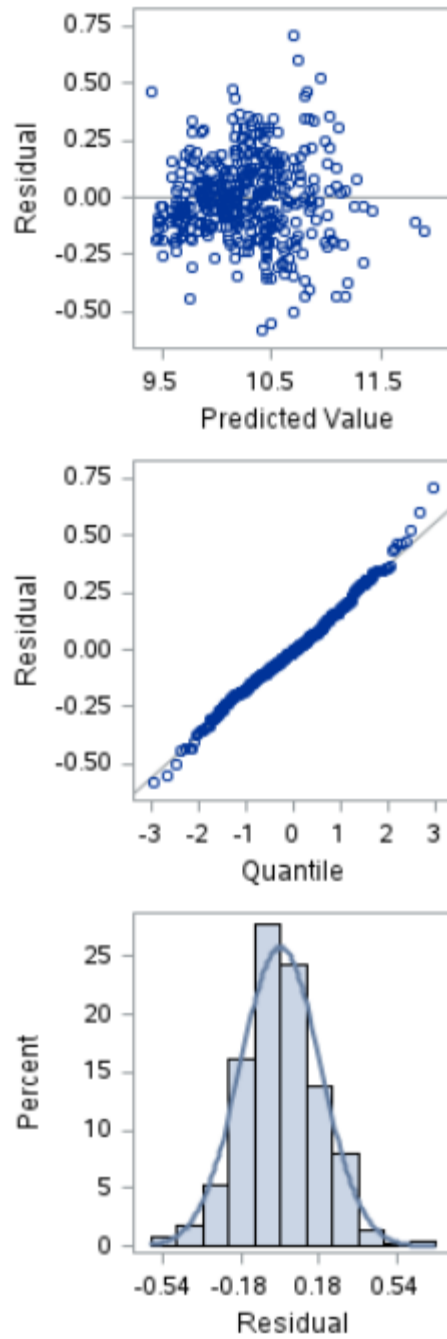


Figure B2:

Figure B5:

The GLM Procedure					
Dependent Variable: log_MSRP					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	16	75.65074606	4.72817163	131.86	<.0001
Error	386	13.84109088	0.03585775		
Corrected Total	402	89.49183694			

R-Square	Coeff Var	Root MSE	log_MSRP Mean
0.845337	1.844849	0.189361	10.26433

Figure B6:

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Vehicle Type	5	0.69954919	0.13990984	3.90	0.0018
Drive Wheels	2	0.83218577	0.41609288	11.60	<.0001
Horsepower*Vehicle Ty	5	1.11548689	0.22309738	6.22	<.0001
Width	1	1.21156834	1.21156834	33.79	<.0001
Weight	1	4.40561187	4.40561187	122.86	<.0001
MPG - City	1	0.31025628	0.31025628	8.65	0.0035
Horsepower	1	1.31248942	1.31248942	36.60	<.0001

Figure B7:

Parameter	Estimate		Standard Error	t Value	Pr > t	95% Confidence Limits	
Intercept	10.09777617	B	0.33598917	30.05	<.0001	9.43717820	10.75837414
Vehicle Type Minivan	0.19328635	B	0.37362475	0.52	0.6052	-0.54130801	0.92788072
Vehicle Type Pickup	-0.17999059	B	0.20641485	-0.87	0.3838	-0.58582877	0.22584759
Vehicle Type SUV	0.03191177	B	0.16052765	0.20	0.8425	-0.28370625	0.34752980
Vehicle Type Sedan	-0.26414682	B	0.12234240	-2.16	0.0315	-0.50468774	-0.02360590
Vehicle Type Sports Car	0.16531353	B	0.15239171	1.08	0.2787	-0.13430820	0.46493526
Vehicle Type Wagon	0.00000000	B	.	.	.	.	.
Drive Wheels AWD	-0.05792633	B	0.03338961	-1.73	0.0836	-0.12357461	0.00772195
Drive Wheels FWD	-0.13776966	B	0.02921241	-4.72	<.0001	-0.19520501	-0.08033431
Drive Wheels RWD	0.00000000	B	.	.	.	.	.
Horsepower*Vehicle Ty Minivan	0.00131126	B	0.00171017	0.77	0.4437	-0.00205115	0.00467368
Horsepower*Vehicle Ty Pickup	0.00146036	B	0.00084195	1.73	0.0836	-0.00019502	0.00311574
Horsepower*Vehicle Ty SUV	0.00177885	B	0.00057815	3.08	0.0022	0.00064213	0.00291558
Horsepower*Vehicle Ty Sedan	0.00429564	B	0.00032514	13.21	<.0001	0.00365636	0.00493491
Horsepower*Vehicle Ty Sports Car	0.00337523	B	0.00036905	9.15	<.0001	0.00264964	0.00410082
Horsepower*Vehicle Ty Wagon	0.00276629	B	0.00060510	4.57	<.0001	0.00157659	0.00395598
Width	-0.02993555		0.00514997	-5.81	<.0001	-0.04006106	-0.01981005
Weight	0.00045214		0.00004079	11.08	<.0001	0.00037194	0.00053234
MPG - City	0.00919366		0.00312550	2.94	0.0035	0.00304852	0.01533880
Horsepower	0.00000000	B	.	.	.	.	.